Panel MS-AR(1) with random-effects intercepts, full sample

Real GDP per worker (period-average USD)

1 Model

Joint panel Markov-switching AR(1) around a *linear* trend, with three regimes:

$$y_{it} = a_i + g\,t + z_{it}, \tag{1}$$
$$z_{i,t+1} = \big(1 - \rho(s_{it})\big)\mu(s_{it}) + \rho(s_{it})\,z_{it} + \sigma(s_{it})\,\varepsilon_{it}, \tag{2}$$
$$s_{i,t+1} \sim \Pi(\,\cdot \mid s_{it}), \tag{3}$$
$$a_i \sim \mathcal{N}(\alpha, \omega^2) \quad \text{(random effects)}. \tag{4}$$

The outcome is log real GDP per worker (period-average USD) from the 2026-09-11 quarterly panel. Calendar time is taken from the period stamp t (year-fraction). The trend is linear in calendar time t (g per year, common across countries). In the pooled specification $a_i \equiv a$. In the random-effects specification the a_i are i.i.d. Normal draws; α and ω are estimated by maximum likelihood, integrating each country's Hamilton-filter likelihood by 11-point Gauss–Hermite quadrature. Cycles are plotted using the posterior mean of a_i. Latent paths s_{it} are country-specific. The regime dated t governs the transition from z_t to z_{t+1}. σ and ρ switch with the regime. The median μ is pinned at 0. $|\rho| < 0.99$. Standard errors (in parentheses) are delta-method from a numerical Hessian on shared parameters. π is the ergodic distribution of Π. $E[z]$ is the long-run mean of the cycle implied by μ, ρ, and Π.

2 Common intercept and linear trend

2026-09-11 panel, full sample: log real GDP per worker. 95 countries, 8937 observations, 1963-Q1–2026-Q2. Fitted countries: 95. Observations: 8937. Log-likelihood: 11694.19. Ergodic mean of the cycle $E[z] = 0.4891$ (0.0830). Common intercept $a = -5.7220$ (0.0997). Common slope $g = 0.0047$ (0.0018).

Table 1: Regime AR(1), transition matrix, and ergodic π. Columns are regimes.

	(0)	(1)	(2)
μ	-0.2840	0.0000	1.3478
	(0.1097)	—	(0.0778)
σ	0.0972	0.4095	0.0403
	(0.0014)	(0.0161)	(0.0005)
ρ	0.9900	0.6361	0.9900
	(0.0000)	(0.0422)	(0.0000)
$\Pi_{0\cdot}$	0.9870	0.0018	0.0112
	(0.0021)	(0.0009)	(0.0020)
$\Pi_{1\cdot}$	0.0238	0.9455	0.0307
	(0.0100)	(0.0125)	(0.0107)
$\Pi_{2\cdot}$	0.0084	0.0028	0.9888
	(0.0015)	(0.0009)	(0.0015)
π	0.4222	0.0417	0.5361
	(0.0377)	(0.0110)	(0.0374)

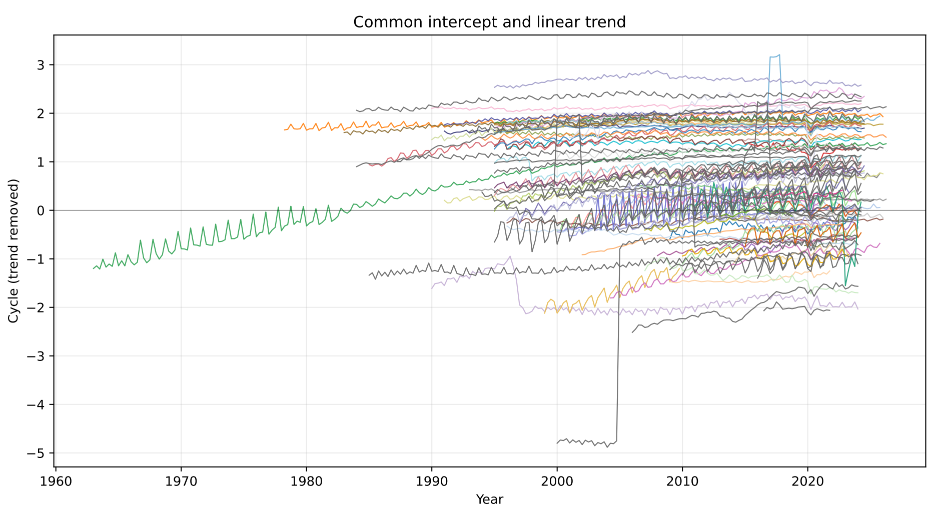

Figure 1: Country cycles after removing the linear trend.

3 Random-effects intercepts, common linear trend

2026-09-11 panel, full sample: log real GDP per worker. 95 countries, 8937 observations, 1963-Q1–2026-Q2. Fitted countries: 95. Observations: 8937. Log-likelihood: 11980.39. Ergodic mean of the cycle $E[z] = 0.0940$ (0.0433). Random intercepts $a_i \sim N(\alpha, \omega^2)$ with $\alpha = -6.3967$ (0.0411), $\omega = 0.7246$ (0.0291). Posterior-mean a_i: mean -6.234, min -7.993, max -4.323. Common slope $g = 0.0208$ (0.0009).

Table 2: Regime AR(1), transition matrix, and ergodic π. Columns are regimes.

	(0)	(1)	(2)
μ	-0.1149	0.0000	0.3166
	(0.1171)	—	(0.0684)
σ	0.0867	0.2599	0.0376
	(0.0025)	(0.0105)	(0.0009)
ρ	0.9898	0.0376	0.9899
	(0.0004)	(0.0622)	(0.0002)
$\Pi_{0\cdot}$	0.9888	0.0047	0.0066
	(0.0024)	(0.0014)	(0.0019)
$\Pi_{1\cdot}$	0.0268	0.9492	0.0239
	(0.0089)	(0.0102)	(0.0087)
$\Pi_{2\cdot}$	0.0052	0.0029	0.9920
	(0.0015)	(0.0010)	(0.0015)
π	0.4036	0.0671	0.5294
	(0.0422)	(0.0145)	(0.0438)

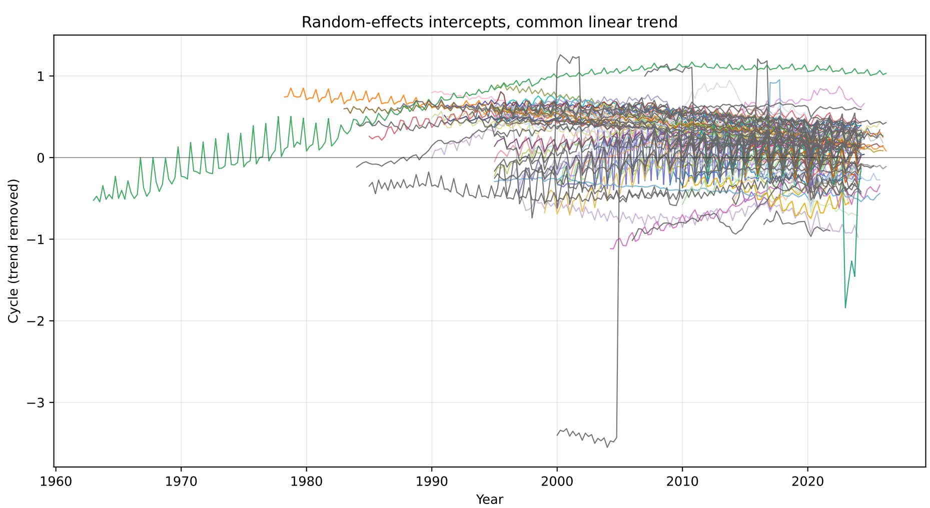

Figure 2: Country cycles after removing the linear trend.